Second Normal Form (2NF)

Introduction

Second Normal Form (2NF) is the next step after First Normal Form (1NF) in database normalization. It focuses on removing **partial dependencies** to ensure that non-key attributes depend on the *whole primary key*.

Definition

A relation (table) is in **Second Normal Form** if:

1. It is already in **First Normal Form (1NF)**.
2. All non-key attributes are **fully functionally dependent** on the *entire* primary key, not just a part of it.

Why 2NF is Important

- Eliminates **partial dependency** problems.
- Reduces redundancy and anomalies during updates.
- Ensures a clearer relationship between attributes and keys.

Example: Violation of 2NF

Consider the following table:

Student_ID	**Course_ID**	**Course_Name**	**Grade**
101	C1	Mathematics	A
101	C2	Physics	B
102	C1	Mathematics	A
102	C3	Chemistry	A

Primary Key: (`Student_ID`, `Course_ID`) — composite key.
Problem:

- `Course_Name` depends *only* on `Course_ID`, not on both `Student_ID` and `Course_ID`.
- This creates a **partial dependency**, violating 2NF.
- Repetition: If a course name changes, it must be updated in multiple rows.

Transforming to 2NF

We separate the data into two tables:

Table 1: Student-Course Grades

Student_ID	Course_ID	Grade
101	C1	A
101	C2	B
102	C1	A
102	C3	A

Table 2: Courses

Course_ID	Course_Name
C1	Mathematics
C2	Physics
C3	Chemistry

Now:

- All non-key attributes depend on the *whole* primary key in Table 1.
- Course name data is stored separately, avoiding redundancy.

Key Points to Remember

- **Partial Dependency:** When a non-key attribute depends on only part of a composite key.
- **2NF Rule:** Remove partial dependencies by creating separate tables.
- If the table has a *single-column primary key*, it automatically satisfies 2NF if it is already in 1NF.

Real-World Analogy

Think of 2NF like **classroom assignments**:

- The student's grade depends on both the student *and* the course (whole key).
- Course details (like name) are stored separately, because they depend only on the course itself.

Summary Table

Rule	Description
Already in 1NF	Table must satisfy all 1NF rules.
No Partial Dependencies	Non-key attributes must depend on the *entire* composite key.
Separate Data	Split tables to store attributes that depend only on part of the key.

Tip

Always check composite keys for partial dependencies. If found, separate the data into multiple related tables using foreign keys.